ODT Summative 2

Idea Brief :

Tetris: Two-Player Edition

What Is It?

A two-player Tetris game built on an ESP32. One player controls the game physically. The other player connects over Wi-Fi on their phone and decides which block comes next.

The Hardware

The device looks like a **Bandai Water Ring Toss toy** , compact and handheld - but with a Tetris screen on the front and **three physical buttons** below it. That's it.

Player 1 - The Player

Sits in front of the device and plays Tetris normally.

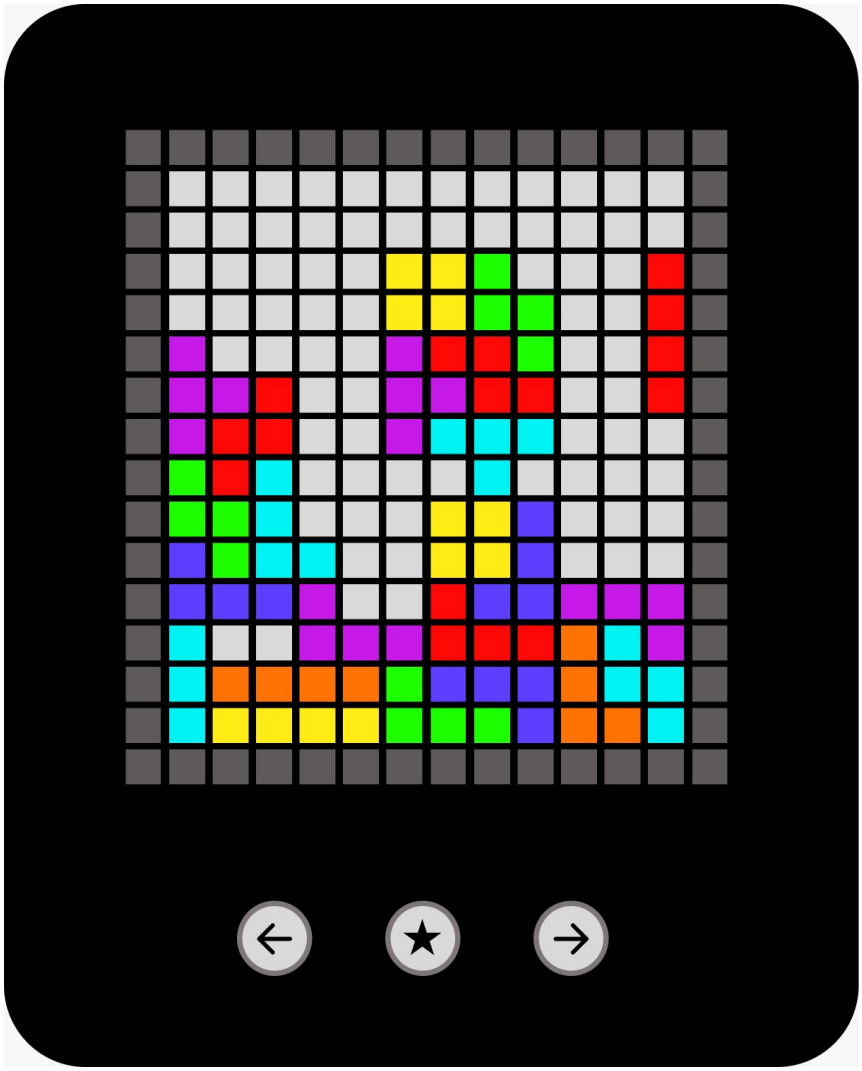
- Press **Play** to start
- Blocks fall down, one row per second
- Use the **Left / Right buttons** to move the block sideways
- Fill a complete row → it disappears, everything above drops down
- Stack reaches the top → Game Over

Player 2 - The Block Picker

Opens a website on their phone (hosted by the ESP32 over Wi-Fi) and picks which block Player 1 gets next.

- Swipe through the 7 Tetris pieces (I, O, T, S, Z, J, L)
- Hit **Select** to queue that piece as the next one to fall
- The screen confirms the choice with a highlight

Why It's Fun

Player 1 is reacting. Player 2 is deciding. They can work together or mess each other up. It turns a solo game into a two-person experience with no app, no internet, just one small device and a Wi-Fi connection.

Basic Idea Sketch :